International Islamic University Chittagong

Department of Electrical and Electronic Engineering
B. Sc. Engineering in EEE
Midterm Exam, Spring 2026

Course Code: **EEE 4707** Course Title: **Power Plant Engineering**

Time: 1 hour 30 minutes Full Marks: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

CO	CO Statement
CO1	Students will be able to understand the operating principles and economics of different power plants.
CO2	They will learn different possible engineering solutions for power plant in societal and environmental contexts
CO3	They will be able to apply mathematical foundations to solve various power plants and power plant economics related issues.

Bloom's Levels of the Questions						
Letter Symbols	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1.	a)	What are the different sources of electrical energy, and how can we derive electricity from these sources.	CO1	C4	**05**
1.	b)	Discuss the challenges of balancing efficiency, cost, and environmental impact in modern power plants. Provide examples of technologies adopted to reduce emissions without sacrificing output.	**CO2**	**C3**	**05**
2.	a)	Compare **steam power plants and gas power plants** in terms of flexibility and sustainability.	**CO1**	**C4**	**05**
2.	b)	A 60 MW steam power station has a thermal efficiency of 30%. If the coal burnt has a calorific value of 6950 kcal/kg, find (i) the coal consumption per kWh, (ii) the coal consumption per day.	CO3	C5	**05**
		Or,			
2.	a)	Explain how balanced draught system works. Differentiate between fire tube boiler and water tube boiler.	**CO1**	**C4**	**05**
2.	b)	A 75 MW steam power station uses coal of calorific value of 6400 kcal/kg. Thermal efficiency of the station is 30% while electrical efficiency is 80%. Calculate the coal consumption per hour when the station is delivering its full output station is delivering its full output.	CO3	C5	**05**
3.	a)	Describe fuel system and lubrication system of Diesel Engine Power Plant with proper diagram.	CO1	C2	**05**
3.	b)	Why is **compressor performance** crucial for overall GTPP efficiency? Briefly explain.	**CO1**	**C4**	**05**

International Islamic University Chittagong

Department of Electrical and Electronic Engineering
B. Sc. Engineering in EEE
Midterm Exam, Autumn 2025

Course Code: **EEE 4707** Course Title: **Power Plant Engineering**

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(i) The figures in the right-hand margin indicate full marks
(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

CO	CO Statement
CO1	Students will be able to understand the operating principles and economics of different power plants.
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Bloom's Levels of the Questions						
Letter Symbols	C1	C2	C3	C4	C5	C6
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1.	a)	Explain why electrical energy is superior to all other forms of energy. Write down the basic principle of power plant.	CO1	C4, C1	05
1.	b)	Discuss how the challenges of balancing efficiency, cost, and environmental factors affect the modern power plants. Provide examples of technologies adopted to reduce emissions without sacrificing output.	CO2	C4	05
2.	a)	Differentiate between fire tube boiler and water tube boiler. How, super-heater, economizer and balanced draught system work in steam power plant?	CO1	C4	05
2.	b)	A 65,000 kW steam power station uses coal of calorific value 15,000 kcal per kg. If the coal consumption per kWh is 0·5 kg and the load factor of the station is 40%, find (i) the overall efficiency, (ii) coal consumption per day.	CO3	C5	05

Or,

2.	a)	Compare jet condensers and surface condensers, highlighting not only the basic difference in their working principle but also their relative merits, limitations, and suitability for modern large-scale power plants.	CO1	C4	05
2.	b)	A coal-fired steam power station is rated at 250 MW peak capacity. The fuel supplied has a calorific value of 7500 kcal/kg. The plant exhibits an electrical efficiency of 91% and an overall thermal efficiency of 24%. - Derive the expression for the total hourly fuel consumption required to sustain the rated output, clearly showing the role of both thermal and electrical efficiencies in the calculation. -Compute the quantity of coal consumed per hour under these operating conditions. -Critically evaluate how variations in efficiency directly affect the fuel requirement of the station.	CO3	C5	05

3.	a)	Describe fuel system and lubrication system of Diesel Engine Power Plant with proper diagram.	CO1	C2	05
3.	b)	With a clearly labeled schematic, illustrate the layout of a Gas Turbine Power Plant (GTPP). Explain in detail the working principle of each major component, analyze the thermodynamic cycle involved, and discuss the factors affecting overall plant efficiency and performance under varying load conditions.	CO1	C6	05

International Islamic University Chittagong

Department of Electrical and Electronic Engineering

B.Sc. Engineering in EEE

Midterm Examination, **Spring-2025**

Course Code: **EEE 4707** Course Title: **Power Plant Engineering**

Time: 1 hour 30 minutes Full Marks: 30

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.

(ii) Course Outcomes (COs) and Bloom's Levels (BL) are mentioned in additional Columns.

Course Outcomes (COs), Program Outcomes (POs) and Bloom's Levels (BL) of the Questions	
CO	**CO Statements**
CO1	Students will be able to understand the operating principles and economics of different power plants.
CO2	They will learn different possible engineering solutions for power plant in societal and environmental contexts.
CO3	They will be able to apply mathematical foundations to solve various power plants and power plant economics related issues.

Bloom's Levels (BL) of the Questions						
Letter Symbols	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1	a)	**(i)** Explain - How electric energy is superior to other form of energy? Also, briefly discuss its importance in the Chattogram region. **(ii)** Explain in your own words the fundamental working principle, benefits, and drawbacks of a steam power station.	**CO1**	**C2**	**3+4**
1	b)	What are the environmental impacts of quick rental power plants? Compare their cleanliness with water, fuel, and nuclear energy from an environmental perspective.	**CO2**	**C3**	**3**
2	a)	**(i)** Draw the schematic arrangement of Automatic Boiler Control System of Steam Power Plant (SPP).Why the automatic boiler control system is important for SPP? **(ii)** What is the function of Boiler furnace and Economizer? Show the classification of Super-Heater. What types of Superheater is commonly used these days and why? In addition, write down an advantage of Steam turbine over Steam engine.	**CO1**	**C1, C4**	**3+4**
2	b)	A steam power station spends Tk. 30 lakhs per annum for coal used in the station. The coal has a calorific value of 5000 kcal/kg and costs Tk. 300 per ton. If the station has thermal efficiency of 33% and electrical efficiency of 90%, find the average load on the station. In addition, mention an way of increasing thermal efficiency of a steam power plant	**CO3**	**C5**	**3**

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OR

2	a)	(i) What is the purpose of steam condenser? Name the types of steam condensers and also differentiate between them. (ii) What is the essential function of pulverizing mills? Show the classification of steam boilers. What types of boiler is used for large capacity steam power stations and why? In addition, mention the function of Super-Heater.	CO1	C1, C4	3+4
2	b)	A 250 MW steam power station operates using coal with a calorific value of 7500 kcal/kg. With an electrical efficiency of 91% and a thermal efficiency of 24%, determine the hourly coal consumption needed to sustain its peak power output. How does the efficiency impact the fuel requirement, and what measures could improve fuel utilization?	CO3	C5	3
3	a)	(i) Illustrate and describe the schematic layout of a Gas Turbine Power Plant (GTPP). Why motor is mounted on the same shaft as that of the turbine of GTPP. (ii) In case of GTPP, what is the significance of Air ratio and Work ratio?	CO1	C4	4+1
3	b)	Describe fuel system and lubrication system of Diesel Engine Power Plant with proper diagram. Also, mention one advantage and disadvantage of such kind of power plant.	CO1	C4	5

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International Islamic University Chittagong

Department of Electrical and Electronic Engineering
B.Sc. Engineering in EEE
Midterm Exam, Spring 2024

Course Code: **EEE 4707** Course Title: **Power Plant Engineering**

Time: 1 hour 30 minutes Full Marks: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

CO	CO Statement
CO1	Students will be able to understand the operating principles and economics of different power plants.
CO2	They will learn different possible engineering solutions for power plant in societal and environmental contexts
CO3	They will be able to apply mathematical foundations to solve various power plants and power plant economics related issues.

Bloom's Levels of the Questions						
Letter Symbols	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1.	a)	Describe the concept of electric energy and its significance in South Asia.	CO1	C1	05
1.	b)	What are the environmental effects of quick rental power plants? Discuss the environmental comparison aspects from cleanliness perspective for Water, Fuel and Nuclear Energy.	CO2	C3	05
2.	a)	Write in own words: the basic principle, advantages, and disadvantages of Steam power station.	CO1	C1	05
2.	b)	Investigate the efficiency of a steam power station, which stands at (57X + 28Y)%. With (4X + 7Y) kg of coal consumed per kWh of electricity generated, determine and elucidate the calorific value of the fuel. X and Y represent the last two digits of your student ID	CO3	C5	05

OR

2.	a)	Write in own words: the comparison between jet and surface condensers.	CO1	C1	05
2.	b)	Consider a 150 MW steam station utilizing coal with a calorific value of 6500 kcal/kg. Given an electrical efficiency of 93% and a thermal efficiency of 25%, calculate the coal consumption per hour required for the station to deliver its maximum rated output.	CO3	C5	05
3.	a)	Describe the lubrication system of Diesel Power Plant with proper diagram.	CO1	C1	05
3.	b)	Draw and explain the schematic arrangement of Gas Turbine Power Plant (GTPP).	CO1	C4	05

International Islamic University Chittagong

Department of Electrical and Electronic Engineering

B. Sc. Engineering in EEE

Midterm Exam, Spring 2023

Course Code: **EEE 4707** Course Title: **Power Plant Engineering**

Time: 1 hour 30 minutes Full Marks: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

	Course Outcomes (COs) of the Questions	PO
CO1	Students will be able to understand the operating principles and economics of different power plants.	POa
CO2	They will learn different possible engineering solutions for power plant in societal and environmental contexts	POg
CO3	They will be able to apply mathematical foundations to solve various power plants and power plant economics related issues.	POb

Bloom's Levels of the Questions						
Letter Symbols	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1.	a)	Differentiate among Power system, Power Station and Power Plant. Point out the importance of electrical energy.	CO1	C4	4+2
1.	b)	What are the sources of energy? Which source you prefer most for producing power in Bangladesh and why?	CO2	C2, C4	1+3
2.	a)	What is the purpose of coal and ash handling plant in Steam power plant? What are the advantages of pulverized coal in steam power plant?	CO1	C1, C1	3+2
	b)	A 150 MW steam station uses coal of calorific value 6500kcal/kg. Electrical efficiency is 93%, thermal efficiency is 25%. Calculate the coal consumption per hour when the station is delivering its full rated output.	CO3	C5	5

Or,

2.	a)	What is meant by Draught in steam power plant? Describe a balanced draught system with its significance.	CO1	C1; C1	2+3
2.	b)	A steam power station has an overall efficiency of (2X+3Y)% and (0.5X+0.5Y) kg of coal is burnt per kWh of electrical energy generation. Calculate and then explain the calorific value of fuel. Where X and Y are last two digits of your ID.	CO3	C5	5
3.	a)	Describe the lubrication system of Diesel Power Plant with proper diagram.	CO1	C1	5
3.	b)	Draw and explain the schematic arrangement of Gas Turbine Power Plant (GTPP).	CO1	C4	5

International Islamic University Chittagong

Department of Electrical and Electronic Engineering

B. Sc. Engineering in EEE

Midterm Examination, Autumn 2022

Course Code: **EEE 4707** Course Title: **Power Plant Engineering**

Time: 1 hour 30 minutes Full Marks: 30

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.

(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Outcomes (COs) of the Questions	
CO1	Students will be able to understand the operating principles and economics of different power plants.
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Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1)	a)	Explain - How electric energy is superior to other form of energy?	CO1	R	5
1)	b)	What are the environmental effects of quick rental power plants? Discuss the environmental comparison aspects from cleanliness perspective of view for Water, Fuel and Nuclear Energy.	CO2	Ap	5
2)	a)	Draw the schematic arrangement of Automatic Boiler Control System of Steam Power Plant (SPP). Why the automatic boiler control system is important for SPP?	CO1	R, An	5
2)	b)	A steam power station spends Tk. 30 lakhs per annum for coal used in the station. The coal has a calorific value of 5000 kcal/kg and costs Tk. 300 per ton. If the station has thermal efficiency of 33% and electrical efficiency of 90%, find the average load on the station.	CO3	E	5
		OR			
2)	a)	What is the purpose of steam condenser? Name the types of steam condensers and also differentiate between them.	CO1	R, An	5
2)	b)	A 150 MW steam station uses coal of calorific value 6500 kcal/kg. Thermal efficiency of the station is 25% and electrical efficiency is 93%. Find the coal consumption per hour when the station is delivering its full rated output.	CO3	E	5
3)	a)	Describe the basic operation principle of Gas turbine power plant (GTPP). Why electric motor is mounted on the same shaft as that of the turbine of GTTP.	CO1	U, An	5
3)	b)	Describe the lubrication system of Diesel Engine Power Plant with proper diagram. Write some advantages of Diesel Engine power plant.	CO1	R	5

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Bloom's Levels of the Questions						
Letter Symbols	**R**	**U**	**Ap**	**An**	**E**	**C**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1)	a)	Differentiate among Power system, Power Station and Power Plant. Point out the importance of electrical energy.	CO1	An	5
1)	b)	What are the environmental effect of quick rental power plants? Discuss the environmental comparison aspects from cleanliness perspective of view for Water, Fuel and Nuclear Energy.	CO2	U	5
2)	a)	Draw the schematic arrangement of Steam Power Plant. Write the basic operation principle of Steam Power Plant.	CO1	R	5
2)	b)	A thermal station has following data: maximum demand=20,000KW, Load factor 40%; Boiler Efficiency 85%; Coal consumption 0.9Kg/Kwh; and cost of coal 300TK/ton. Determine: (i) Thermal Efficiency (ii) Coal bill per year.	CO3	E	5

Or,

2)	a)	How coal and ash is handled in Steam power plant? What are the advantages of pulverized coal in steam power plant?	CO1	R	5
2)	b)	A coal station has annual coal consumption bill 1,90,00,000 TK. Coal consumption 0.9 kg/kwh. If Load factor =40% and cost of coal 300TK/ton. Determine: (i) Average load and (ii) maximum demand.	CO3	E	5
3)	a)	Name one of the gas based power plants in Bangladesh with it's capacity. Describe the operation of close cycle gas turbine with proper diagram.	CO1	R	5
3)	b)	Draw and explain the schematic arrangement of Gas Turbine Power Plant (GTPP).	CO1	An	5